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Enhancing Aircraft Safety through Explainable AI-Based Failure Detection Systems

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ABSTRACT

Modern aircraft systems have evolved into complex cyber-physical architectures equipped with thousands of sensors that generate terabytes of operational data during every flight. Extracting meaningful insights from this data to ensure safety, reliability, and optimal performance has become an increasingly critical challenge. Artificial Intelligence (AI) and machine learning techniques are now widely applied in predictive maintenance and fault detection; however, the inability to interpret or justify their decisions poses a significant barrier to real-world adoption in safety-critical aviation environments.

This paper proposes an Explainable Artificial Intelligence (XAI) framework for aircraft failure diagnosis and decision support, focusing on making AI-driven predictions transparent, interpretable, and actionable. The proposed approach combines deep neural networks (DNNs) for fault classification with explainability layers such as SHapley Additive exPlanations (SHAP) and Local Interpretable Model-Agnostic Explanations (LIME) to provide interpretable justifications for every diagnostic outcome. Experimental validation using the NASA C-MAPSS Turbofan Engine dataset demonstrates the framework's capability to achieve high diagnostic accuracy (95.6%) while offering interpretable visual explanations. The findings highlight that explainable AI not only enhances the trust of maintenance engineers and flight safety officers but also reduces decision latency and operational downtime.

Keywords: Explainable Artificial Intelligence (XAI), Aircraft Maintenance, Predictive Fault Diagnosis, Decision Support Systems, SHAP, LIME, Deep Learning, Human-in-the-Loop.

I. Introduction

The aviation industry relies on systems that must operate flawlessly under extreme environmental conditions. Even minor equipment malfunctions can lead to significant safety risks, unscheduled maintenance events, and financial losses. Predictive maintenance powered by artificial intelligence (AI) has emerged as a powerful solution to anticipate such failures before they escalate. However, traditional AI models—though effective in accuracy—often operate as “black boxes,” providing predictions without insight into their internal reasoning.

This opacity raises substantial challenges in aviation, where regulatory compliance and human accountability are non-negotiable. For example, if an AI model predicts an engine failure but fails to justify its decision, maintenance engineers cannot take corrective actions with confidence. Therefore, explainability has become a fundamental requirement in AI systems used for aircraft health monitoring and decision support.

Explainable AI (XAI) seeks to make the reasoning behind AI models human-interpretable by identifying the most influential parameters in a prediction, visualizing their relationships, and translating them into actionable insights. By providing a transparent layer of interpretation, XAI bridges the gap between model accuracy and human trust—ensuring that AI serves as a reliable co-pilot rather than a mysterious automation engine. This research focuses on developing and evaluating a hybrid XAI-based framework for aircraft failure diagnosis that balances predictive accuracy with interpretability, facilitating better decision-making by both maintenance teams and flight safety officers.

II. Related Work

Early approaches to aircraft fault diagnosis primarily relied on rule-based expert systems and model-based reasoning. These systems leveraged predefined rules or mathematical models representing system dynamics. However, they lacked adaptability to changing operational conditions and sensor noise.

The advent of machine learning revolutionized this domain. Techniques such as Support Vector Machines (SVM), Decision Trees (DT), and Random Forests (RF) were employed for pattern recognition in sensor data. Later, deep learning architectures such as Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks demonstrated exceptional performance in identifying nonlinear patterns in multivariate time-series data.

However, interpretability remained a persistent challenge. Recent literature has begun to focus on explainability. While these works have demonstrated promising results, most focus on isolated subsystems and lack an integrated decision-support interface. This research addresses this gap by developing a comprehensive hybrid framework combining interpretability, visualization, and human-centered decision-making for holistic aircraft fault analysis.

III. Proposed Architecture

The proposed Explainable AI Framework for Aircraft Failure Diagnosis consists of four primary modules as illustrated in Table 1.

Module	Description	Key Algorithms/Techniques Used	Output/Result
Data Acquisition & Preprocessing	Collects real-time sensor data (temperature, vibration, pressure, torque, fuel flow, etc.) and cleans anomalies.	Data normalization, feature scaling, mutual information-based feature selection	Structured sensor data ready for analysis
Fault Detection & Classification	Identifies fault patterns and classifies failure types.	Deep Neural Network (DNN), Gradient Boosting Trees (GBT)	Fault classification with high accuracy
Explainability Layer (Post-hoc XAI)	Generates interpretable justifications for model outputs.	SHAP and LIME	Visual explanations for model predictions
Decision Support Interface	Provides engineers with actionable insights and recommendations.	Interactive dashboard, visual analytics, confidence scoring	Real-time decision aid and fault localization

IV. Methodology

The methodology followed in this study includes five main stages: data acquisition, preprocessing, model development, explainability integration, and evaluation. The experiments used NASA's C-MAPSS Turbofan Engine Degradation dataset, simulating realistic engine wear and fault progression. The dataset contains 21 sensor readings over multiple cycles.

Data preprocessing involved outlier removal, Z-score normalization, and Principal Component Analysis (PCA) to reduce noise while retaining 98% variance. Missing values were handled using K-Nearest Neighbor (KNN) interpolation.

The Deep Neural Network (DNN) architecture used three hidden layers with 256, 128, and 64 neurons respectively. The model was trained using the Adam optimizer, with categorical cross-entropy loss. Gradient Boosting Trees were trained in parallel for global feature importance validation.

Post-training, SHAP and LIME algorithms were applied to explain model decisions. SHAP provided global interpretability and LIME focused on local explanations. A custom-built dashboard integrated with Plotly and Dash allowed visualization of these explanations in real-time.

Model performance was evaluated using accuracy, precision, recall, and F1-score. Mean Explanation Fidelity (MEF) and Human Interpretability Score (HIS) were used as explainability metrics, with human-in-the-loop evaluations conducted by aerospace engineers.

V. Experimental Results and Discussion

The proposed DNN-based fault diagnosis system achieved a diagnostic accuracy of 95.6%, outperforming traditional baselines such as Random Forest (91.7%) and SVM (89.3%). The F1-score was recorded at 0.94, demonstrating robust classification consistency.

Feature analysis using SHAP revealed that compressor outlet temperature, fan speed, and fuel flow were the most influential factors in predicting turbine faults. LIME visualizations confirmed that abnormal exhaust gas temperatures strongly correlated with early degradation stages.

A human-centered evaluation showed that aerospace engineers' decision verification time decreased by 37%, and their trust level in AI predictions increased by 41%. The Explainability Fidelity Score (0.92) validated the system's interpretability alignment with actual physical parameters. SHAP

dependency plots revealed nonlinear interactions between compressor pressure ratio and fuel temperature—insights previously unnoticed through manual analysis.

VI. Future Work

Future developments will explore temporal explainability to interpret degradation patterns over time using attention-based sequence models. Causal reasoning and knowledge graphs will be integrated to move beyond correlation-based explanations and towards cause-effect analysis. The framework will be deployed in real-time flight data environments through edge computing for on-board explainability. Compliance with EASA and FAA standards will also be targeted, ensuring regulatory transparency. Finally, incorporating human feedback loops will refine AI decision-making over continuous operational cycles.

VII. Conclusion

This research establishes the importance of Explainable AI (XAI) in aircraft failure diagnosis and decision support. By combining deep learning with interpretable explanation methods such as SHAP and LIME, the system achieved high diagnostic accuracy while offering transparent justifications for each prediction. This fosters greater trust and understanding among maintenance engineers and flight safety managers. The integration of human feedback and explainability ensures that AI-driven systems not only predict failures accurately but also provide clarity and accountability—critical for aviation safety.

As aviation systems continue to evolve, XAI will play a central role in developing trustworthy, regulation-compliant, and human-centric AI applications that redefine the boundaries of intelligent aircraft maintenance.

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